YO Mock Orals 2025

I’m providing a transcript or notes from a mock oral therapeutics case.
Please clean up the content, correct medical terminology, and create structured, exam-ready notes in the same format as my previous *Acute Anterior Uveitis* document.

Formatting instructions:

- Use Google Docs–friendly **dot points** and **indents** (no page or section line breakers).
- Follow this **section order**:
    1. Case Overview
    2. History Questions
    3. Symptoms (Subjective Findings)
    4. Entrance Tests (VA, IOP, Motilities, Pupils)
    5. Slit-Lamp / Anterior Segment Findings
    6. Posterior Segment (if relevant)
    7. Diagnosis and Differentials
    8. Management (with reasoning and key dosing details)
    9. Review, Prognosis, and Tapering
    10. Referral and Red Flags
    11. Review of Systems or Key Associated Conditions (if relevant)
- Include 💭 **reasoning prompts** throughout for oral viva recall.
- Add **summary tables** (grading scales, medication comparisons, or systemic associations) wherever useful.
- Keep tone concise but clinically detailed (as per university oral therapeutics standard).
- Use **Australian English**.
- Do **not** include page line dividers or markdown formatting.

Acute Anterior Uveitis (Non-Granulomatous) — Oral Exam Notes

1. Case Overview

- 25-year-old male presented with increasing redness and pain in the right eye for 3 days.
- Wore sunglasses in consult (photophobic).
- No trauma, surgery, contact lens wear, systemic illness, or medications.
- No allergies, discharge, or itch.
- VA 6/7- OD, 6/6 OS.
- Impression: unilateral painful photophobic red eye → Acute Anterior Uveitis (AAU).

2. History Questions

- Onset and duration of redness/pain?
- Nature of pain (dull ache vs sharp)? Does light worsen it?
- Any discharge, itch, or crusting?
- History of trauma or swimming pool exposure?
- Any vision changes?
- Previous similar episodes?
- Joint pain, back stiffness, rashes, mouth/genital ulcers, bowel or urinary symptoms?
- Any autoimmune disease or recent infection
- Medications or family history of arthritis, psoriasis, IBD?
  💭 Purpose: to screen for systemic causes of uveitis (e.g. HLA-B27, sarcoid, TB, syphilis, JIA, Behçet).

3. Symptoms

- Dull pain 5/10 with significant photophobia.
- Mild blur in right eye.
- No discharge or itch.
- Systemically well.
  💭 Pattern: Pain + photophobia + blur + no discharge → inflammatory, not infective.

4. Entrance Tests

- Visual Acuity: OD 6/7–, OS 6/6.
  💭 Larger VA drop = posterior involvement suspicion.
- IOP: OD 16 mmHg, OS 16 mmHg.
  💭 Early = low (↓ aqueous production), late = high (steroid responder).
- Motilities: Full, pain-free.
  💭 Pain on movement = scleritis/orbital issue.
- Pupils: OD sluggish, slightly mid-dilated, irregular margin; OS normal.
  💭 Suggests iris sphincter spasm and early posterior synechiae.

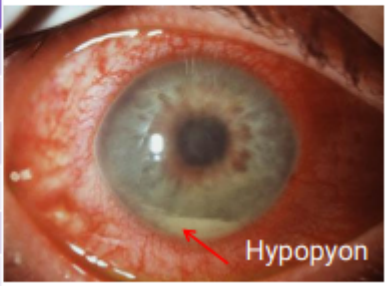
5. Slit-Lamp / Anterior Segment Findings

- Ciliary flush around limbus.
- Cornea clear, no epithelial defect or infiltrates.
- Endothelium: fine stellate KPs (non-granulomatous).
  💭 Granulomatous = large “mutton-fat” greasy KPs.
- Anterior chamber: fibrinous flare (Grade 4+) and ~2+ cells (adjusted for 1×1 mm beam).
- Very mild hypopyon inferiorly.
- Iris irregular with posterior synechiae starting.

- Corneal sensation normal → not herpetic.

Grading anterior chamber cells & flare

Grade	Cells in field	Flare (description)
0	<1	None
0.5	1 – 5	None
1	6 – 15	Faint
2	16 – 25	Moderate (iris and lens details clear)
3	26 – 50	Marked (iris and lens details hazy)
4	> 50	Intense (fibrin or plastic aqueous)



Hypopyon Overview

Feature	Explanation
Definition	Layering of inflammatory cells + fibrin in AC inferiorly
Common in	Severe anterior uveitis (e.g. HLA-B27, Behçet, endophthalmitis)
Appearance	White/yellow fluid level, non-shifting if sterile
Management	Treat underlying inflammation (steroid + cycloplegic) — no separate treatment required
When to refer	Large, purulent, or shifting hypopyon → suspect microbial keratitis/endophthalmitis

6. Posterior Segment

- Always dilate to rule out intermediate/posterior uveitis.
- Look for vitritis, snowballs, snow-banking, vasculitis, or CME.
- If present → immediate ophthalmology referral (within 24 h).
- Expected here: clear fundus (isolated anterior disease).

7. Diagnosis and Differentials

- **Diagnosis:** Acute Non-Granulomatous Anterior Uveitis.
- **Differentials:**
  - Keratitis (HSV, microbial) → epithelial defect/dendrite.
  - Acute angle-closure glaucoma → severe pain, corneal oedema, ↑IOP.
  - Conjunctivitis → discharge, itch, no photophobia.
  - Scleritis → deep pain, vessels non-blanching.
  - Intermediate/posterior uveitis → floaters, mild redness.

8. Management

- **Topical Corticosteroid – Prednefrin Forte 1% (Prednisolone acetate):**
  - High penetration and potency.
  - PBS-restricted for uveitis → cost-effective.
  - Loading dose: every 15–30 min for 2 h → hourly for 24 h.
  - Continue hourly until ≥ 2-step improvement in SUN grade.
  - Maintenance: hourly → QID once quiet.
  - At night: 3 drops before bed or Hycor 1% ointment.
    - 💭 Why pred acetate? → best uveal penetration.
    - 💭 Contraindicated: active herpetic keratitis, pregnancy, known steroid responder.
    - 💭 Alternative: Maxidex (dexamethasone) → stronger but ↑ IOP, irritation.
- **Cycloplegic – Atropine 1% BD–QID:**
  - Relieves ciliary spasm & pain.
  - Prevents posterior synechiae formation.
  - Stabilises blood–aqueous barrier.
  - Stop when pain resolved and pupil mobile.
    - 💭 If omitted: persistent pain, synechiae, irregular pupil, secondary glaucoma.
- **If synechiae persist:**
  - Add Phenylephrine 2.5% in-office (safe).
  - If no release, 10% with BP check (normal ≈ 120/80).
  - Avoid in hypertension or cardiac disease.
- **IOP monitoring & control**
  - Check at each review.
  - Steroid response: 2–6 weeks post-start.
  - If ↑IOP → add Timolol or Brimonidine (avoid PGAs).

9. Review, Prognosis, and Tapering

- First review at 24 h → assess pain, AC activity, IOP.
- Then every 2–3 days until quiet.
- Resolution: usually 2–6 weeks.
- Quick taper: QID × 4 d → TDS × 3 d → BD × 2 d → OD × 1 d → alternate days × 1 → stop.
- Atropine stopped once pain subsides.
- Hypopyon resolves with inflammation control (no separate Rx).
- Resolution = Grade 0 cells/flare, normal IOP, comfortable eye.
  💭 *Steroid response = IOP ↑ > 5 mmHg.*

10. Referral and Red Flags

- Immediate referral (≤ 24 h) if:
  - Intermediate/posterior involvement.
  - Granulomatous, bilateral, or recurrent cases.
  - Severe fibrinous reaction or fixed hypopyon.
  - Post-surgical onset.
  - Pregnancy or uncertain diagnosis.
- Otherwise manage collaboratively and investigate systemic associations.

11. Review of Systems — Systemic Associations

System	Symptoms	Associated Conditions
General	Fever, weight loss, malaise, night sweats	Inflammatory Bowel Disease (IBD), Tuberculosis (TB), Systemic Lupus Erythematosus (SLE)
Rheumatologic	Arthralgias, lower back pain, joint stiffness	Juvenile Idiopathic Arthritis (JIA), Ankylosing Spondylitis, Reactive Arthritis, HLA-B27 Antigen, Systemic Lupus Erythematosus (SLE)
Dermatologic	Rashes, sores, alopecia, vitiligo, poliosis, tick/insect bites	Lyme Disease, Systemic Lupus Erythematosus (SLE)
Neurologic	Tinnitus, headaches, meningism, paresthesias, weakness/paralysis	Multiple Sclerosis (MS), Vogt-Koyanagi-Harada Syndrome
Respiratory	Shortness of breath, cough, sputum production	Tuberculosis (TB), Sarcoidosis
Gastrointestinal (GI)	Diarrhea, bloody stools, oral aphthous ulcers	Behçet's Disease, Inflammatory Bowel Disease (IBD)
Genitourinary (GU)	Dysuria, discharge, genital ulcers, balanitis, sexually transmitted infections (STIs)	Syphilis, Behçet's Disease

Key Oral-Exam Phrases

- "Fine stellate KPs with 2+ cells and 4+ fibrinous flare consistent with non-granulomatous anterior uveitis."
- "Pred acetate 1% hourly with loading dose; atropine 1% BD; review in 24 hours."
- "Hypopyon is mild and sterile; resolves with anti-inflammatory therapy."
- "Monitor IOP and taper once quiet; avoid prostaglandin analogues."
- "Refer immediately if posterior, bilateral, or granulomatous involvement."

PIGMENT DISPERSION SYNDROME (PDS) & PIGMENTARY GLAUCOMA (PG)

Case Overview

- Mock image: heavily pigmented open angles on gonioscopy; Krukenberg spindle on corneal endothelium; mid-peripheral iris transillumination defects (TIDs); posterior iris bowing
    - Very high IOP in both eyes (≈35+ mmHg); VA unaffected
    - Younger myopic male CAUASIAN

- Pattern consistent with **PDS**, with risk of progression to **PG** if structural/functional damage is present
- 💭 Reasoning: PDS is a secondary open-angle mechanism—iris–zonule friction liberates pigment ➜ TM blockage ➜ IOP spikes

History Questions

- General glaucoma risks: family history; long-term steroid use; trauma; migraine/vascular dysregulation; sleep apnoea
- PDS-focused: episodes of blur/haloes after **exercise** or **dilation**; myopia; past IOP spikes; strenuous activity patterns
- Contraindications checklist for drop choice: asthma/COPD, bradycardia/heart block, depression, pregnancy/breastfeeding, sulfonamide allergy, renal/hepatic disease
- 💭 Reasoning: identify secondary causes, pick safe therapy (drops vs SLT), anticipate adherence issues

Symptoms (Subjective Findings)

- Often asymptomatic early
  - Intermittent blur/haloes (post-exercise IOP spikes)
  - Mild ache/photophobia possible
- 💭 Reasoning: transient spikes map to pigment “showers” with exertion or dilation

Entrance Tests (VA, IOP, Motilities, Pupils)

- Visual acuity: typically normal at presentation
- IOP: elevated or fluctuating; check both eyes and consider diurnal variation if suspicious
- Motilities: full; cover test baseline for fields later
- Pupils: RAPD only if advanced damage
- 💭 Reasoning: high IOP without disc/field loss = OHT; with loss = glaucoma

Severity (IOP band)	Typical range	Urgency / initial action
Mild	< 22 mmHg	Routine review; risk-factor assessment
Moderate	22–29 mmHg	Prompt review; confirm with repeat IOP/fields/OCT; consider treatment
Severe	≥ 30 mmHg	**Urgent** initiation/adjustment of therapy; close follow-up

Slit-Lamp / Anterior Segment Findings

- Cornea: **Krukenberg spindle** (vertical endothelial pigment)
- Iris: **mid-peripheral TIDs**; **posterior iris bowing** (“reverse pupillary block”)
- Lens: posterior capsule pigment granules may be seen
- Angle (gonioscopy): 360° **heavily pigmented TM** (often worse inferiorly); possible Sampaolesi line; **open, deep** angles

- Anterior chamber: usually quiet (no significant cells/flare)
- 💭 Reasoning: posterior bowing increases iris–zonule contact → pigment liberation → TM clogging → fluctuating IOP rise

Posterior Segment (if relevant)

- Optic disc: vertical C/D asymmetry, rim thinning, notching, disc haemorrhage
- OCT RNFL/GCC: focal or diffuse thinning
- Visual fields: nasal step/arcuate defects matching OCT pattern
- 💭 Reasoning: confirm structural–functional correlation to call **PG** (not just PDS)

Diagnosis and Differentials

- Primary: **Pigment Dispersion Syndrome**; **Pigmentary Glaucoma** if disc/field damage present
- Key differentials:
    - Pseudoexfoliation (PEX) material (older, flaky lens-capsule ring, zonular issues)
    - Steroid-induced OAG (medication history)
    - Chronic uveitis pigment (inflammation history, KPs)
    - Angle recession (trauma)
    - ICE syndrome (endothelial changes)
- Viva sound-bite definitions:
    - *PDS* = secondary OAG mechanism from iris–zonule friction with pigment deposition in outflow pathways
    - *PG* = PDS + glaucomatous optic neuropathy with corresponding field loss
- 💭 Reasoning: age + myopia + posterior bowing + spindle + dense TM pigment → PDS/PG > PEX

Management (with reasoning and key dosing details)

- Goals: lower IOP, reduce fluctuation, protect optic nerve
- First-line choices (pick and justify):
    - **Prostaglandin analogue** once nightly (if not pregnant/uveitic): potent single agent; ↑ uveoscleral outflow
    - **Primary SLT** in open angles—useful when adherence uncertain; in very pigmented angles use **lower energy**, consider **staged treatment**, and schedule **post-laser IOP check** (risk of spike)
- Adjuncts/add-ons:
    - **Beta-blocker** (↓ production), **alpha-2 agonist** (↓ production; ↑ uveoscleral outflow), **topical CAI** (↓ production), **combination** drops
- Lifestyle: avoid vigorous exercise immediately after dilation; address sleep apnoea; avoid unnecessary steroids
- 💭 Reasoning prompts:
    - Which is safer right now—once-nightly PGA or primary SLT—given age, adherence, pregnancy potential, and pigment load?
    - In a heavily pigmented angle, how do you mitigate SLT spike risk (pretreat, lower energy, segmental shots, next-day IOP check)?

Class	Example (concentration)	Typical dose	Mechanism	Key adverse effects / notes

Prostaglandin analogues	Latanoprost 0.005%, Travoprost 0.004%, Bimatoprost 0.01–0.03%	OD nocte	↑ Uveoscleral outflow	Hyperaemia, lash growth, iris/skin pigmentation; **avoid in pregnancy**; avoid in active uveitis
Beta-blocker	Timolol 0.25–0.5%	BD	↓ Aqueous production	Bradycardia, bronchospasm, hypotension; caution asthma/COPD/heart block
Alpha-2 agonist	Brimonidine 0.1–0.2%	BD–TDS	↓ Production; ↑ Uveoscleral outflow	Dry mouth, fatigue, allergy; **avoid near term/infancy**
Topical CAI	Dorzolamide 2%, Brinzolamide 1%	TDS (Dorzo), BD–TDS (Brinzo)	↓ Aqueous production	Stinging/bitter taste; caution sulfonamide allergy
Rho-kinase inhibitor	Netarsudil 0.02%	OD nocte	↑ TM outflow (ROCK/NET inhibition)	Hyperaemia, corneal verticillata; limited pregnancy data
Miotic (select cases)	Pilocarpine 1–2%	QID	↑ Trabecular outflow; reduces reverse pupillary block	Headache, brow ache, myopic shift; rarely long-term in PDS
Laser	SLT	—	Improves TM outflow	Use lower energy in pigmented angles; plan IOP check
Surgery	MIGS / Trab	—	Lowers IOP via outflow bypass/filtration	Reserve for refractory disease

Option in pregnancy	Practical stance (concise)	Notes you can say in viva
Prostaglandin analogues	**Avoid**	Uterotonic class concern; prefer alternatives/SLT
Timolol	**Cautious use**	Consider punctal occlusion; avoid if maternal asthma/COPD/heart block
Brimonidine	**Avoid near term/infancy**	Risk of neonatal CNS depression; specialist advice if considered

Topical CAI	**Limited data**	Reserve if benefit > risk; consider specialist input
Oral CAI (acetazolamide/methazolamide)	**Avoid unless specialist-directed**	Systemic exposure; reserve for sight-threatening scenarios
SLT	**Preferred non-drug option**	Avoids systemic drug exposure in pregnancy
Surgery	**Specialist decision**	For refractory, sight-threatening disease only

Review, Prognosis, and Tapering

- Initial review: **4 weeks** after starting therapy or SLT → IOP, side-effects, adherence
- Ongoing monitoring: **3–6 monthly** if stable; more often if high risk (younger age, very high IOP, disc haemorrhage)
- Targets: set a **low-teens** target if damage present; lower if progression continues
- Prognosis: variable; may improve with age as iris stiffens (less pigment release)
- 💭 Reasoning: if fields/OCT worsen at target, either **lower the target** or **escalate** (add drug, SLT, surgical referral)

Referral and Red Flags

- Co-manage/refer if: IOP ≥ 35 mmHg despite therapy; rapid disc/field progression; medication intolerance
- Emergency if: acute painful spike with nausea/corneal oedema
- Surgical referral: inadequate control on maximally tolerated meds/SLT; consider MIGS/trab per surgeon

Review of Systems or Key Associated Conditions (if relevant)

- Myopia (structural risk)
- Sleep apnoea (nocturnal IOP spikes)
- Vascular dysregulation (migraine, hypotension)
- Steroid exposure (iatrogenic IOP rise)
- 💭 Reasoning: systemic contributors can worsen progression at a given IOP

Theory of glaucoma	Core idea	Exam cues to say
Mechanical	High IOP deforms lamina cribrosa → axonal transport block	“Lamina stress/strain → RGC loss”
Vascular/Ischaemic	ONH hypoperfusion/autoregulation failure	“Disc haemorrhage, nocturnal hypotension”
Cellular/Mitochondrial	Axonal metabolic failure/oxidative stress	“Mitochondrial dysfunction → apoptosis”

PSEUDOEXFOLIATION GLAUCOMA (PXG)

Case Overview

- Image findings: rough, irregular iris margin; concentric white/grey deposits on anterior lens capsule resembling “frosted” or “dandruff-like” flakes
    - Material typically accumulates at the **pupillary margin** and **lens zonules**
    - Gonioscopy: moderate–heavy pigmentation (often inferiorly)
    - VA normal early; IOP variable but often elevated (>30 mmHg)
    - Age: 60–80 years (mean ~70 years); commonly Scandinavian or Caucasian ancestry
- 💭 Reasoning: pseudoexfoliation (PEX) material obstructs trabecular meshwork and weakens zonules → secondary open-angle glaucoma

History Questions

- **Ask about:**
    - Previous cataract or ocular surgery (risk of zonular dialysis)
    - Systemic vascular history (ischaemic disease, hypertension)
    - Family history of glaucoma
    - Trauma or steroid use
    - Ethnic background (higher prevalence in Scandinavian populations)
    - Medication use and contraindications (for IOP-lowering drops)
- 💭 **Reasoning: PXG has systemic associations (vascular, cardiac) and surgical implications (zonular weakness)**

Symptoms (Subjective Findings)

- **Often asymptomatic until advanced**
    - Gradual loss of peripheral vision
    - Possible intermittent blurred vision from IOP spikes
    - No acute pain unless extreme pressure elevation
- 💭 **Reasoning: PXG is chronic, asymmetrical, and progresses faster than POAG once present**

Entrance Tests (VA, IOP, Motilities, Pupils)

- **Visual Acuity:** typically preserved early
- **Intraocular Pressure:** raised, fluctuating; can spike severely (>40 mmHg)
- **Motilities:** full
- **Pupils:** normal reactions unless advanced optic neuropathy
- 💭 Reasoning: large diurnal IOP fluctuations due to TM blockage by PEX debris

Severity (IOP band)	Typical range	Urgency / action
Mild	< 22 mmHg	Routine review; assess risk factors
Moderate	22–29 mmHg	Prompt review; repeat IOP/fields/OCT
Severe	≥ 30 mmHg	Urgent initiation of therapy; close follow-up

Slit-Lamp / Anterior Segment Findings

- Lens capsule: **pseudoexfoliative material** forming a central disc with a **clear intermediate zone** (wiping effect of iris movement) and **peripheral granular zone**
- Iris: **moth-eaten or irregular pupillary ruff**; pigment loss along the margin
- Zonules: may show deposits and fragility (phacodonesis, subluxation)
- Cornea: possible pigment deposition inferiorly
- Angle (gonioscopy): patchy TM pigmentation; possible Sampaolesi line
- 💭 Reasoning: PEX material from abnormal basement membrane metabolism accumulates on anterior segment structures ➔ mechanical blockage and poor outflow

Posterior Segment (if relevant)

- ONH: glaucomatous cupping, rim loss, peripapillary atrophy
- Visual field: more **rapid progression** and **greater asymmetry** than POAG
- 💭 Reasoning: PXG leads to higher mean IOP and worse IOP fluctuation ➔ faster structural and functional loss

Diagnosis and Differentials

- **Primary: Pseudoexfoliation Glaucoma (PXG)**
- **Related conditions:**
  - Pseudoexfoliation Syndrome (PXF) without glaucomatous damage
  - Primary open-angle glaucoma
  - Pigmentary glaucoma (younger, myopic, mid-peripheral TIDs)
  - Uveitic glaucoma (with AC inflammation)
- **Viva definitions:**
  - *Pseudoexfoliation syndrome* – systemic fibrillopathy causing deposition of grey-white material on ocular structures
  - *Pseudoexfoliation glaucoma* – open-angle glaucoma secondary to PEX material obstructing the TM
- 💭 **Reasoning: PXG = most common identifiable cause of secondary OAG worldwide**

Management (with reasoning and key dosing details)

- **Goal:** lower IOP, reduce fluctuation, prevent optic nerve damage
- **First-line:**
  - **Prostaglandin analogue (e.g. Latanoprost 0.005% nocte)** — potent, once daily
  - Alternatively or adjunct: **Beta-blocker** (Timolol 0.25–0.5% BD), **Alpha-2 agonist** (Brimonidine 0.2% BD–TDS), or **Topical CAI** (Dorzolamide 2% TDS / Brinzolamide 1% BD–TDS)
- **Combination therapy:** often required early due to higher baseline IOPs
- **Laser: SLT** or **ALT** effective but response may diminish; may require repeat
- **Surgery:** earlier consideration (trabeculectomy or MIGS) — higher failure risk due to PEX material deposition and post-op inflammation
- 💭 Reasoning prompts:
  - Why treat earlier? ➔ PXG progresses faster and has worse IOP control than POAG
  - What risks with cataract surgery? ➔ Zonular weakness ➔ capsular rupture or lens subluxation
  - How does pigment load affect SLT? ➔ Higher risk of inflammation ➔ use lower energy and divided sessions

Class	Example (concentration)	Typical dose	Mechanism	Key adverse effects / notes
Prostaglandin analogues	Latanoprost 0.005%, Travoprost 0.004%	OD nocte	↑ Uveoscleral outflow	Hyperaemia, lash growth, pigmentation; **avoid in pregnancy**
Beta-blocker	Timolol 0.25–0.5%	BD	↓ Aqueous production	Bradycardia, bronchospasm; avoid in asthma/COPD
Alpha-2 agonist	Brimonidine 0.1–0.2%	BD–TDS	↓ Production; ↑ Outflow	Allergy, fatigue; **avoid near term/infancy**
Topical CAI	Dorzolamide 2%, Brinzolamide 1%	TDS / BD–TDS	↓ Aqueous production	Stinging, metallic taste
ROCK inhibitor	Netarsudil 0.02%	OD nocte	↑ Trabecular outflow	Hyperaemia; limited data
Laser / SLT	—	—	Improves TM outflow	Effect wanes over time; avoid overtreatment
Surgery	Trabeculectomy / MIGS	—	Mechanical outflow	Reserved for refractory PXG

Pregnancy / Treatment option	Clinical stance	Key comment
Prostaglandin analogues	**Avoid**	May trigger uterine contractions
Timolol	**Use cautiously**	Punctal occlusion; avoid with asthma/COPD
Brimonidine	**Avoid near term/infants**	Neonatal CNS depression
Topical CAI	**Limited data**	Reserve if benefit > risk
Oral CAI	**Avoid unless essential**	Teratogenic potential
SLT	**Preferred non-drug**	Reduces systemic exposure
Surgery	**Specialist-directed**	For uncontrolled PXG or combined cataract–glaucoma cases

Review, Prognosis, and Tapering

- Review 4 weeks after initiating or changing therapy
- Reassess IOP, optic disc, fields, OCT, and compliance

- **Progress faster** than POAG; expect more IOP fluctuation and poorer medication response
- **Targets:** often < 16 mmHg, ideally low teens
- **Prognosis:** guarded; requires lifelong monitoring and possible surgical escalation
- 💭 Reasoning: IOP spikes and zonular fragility make control difficult—early identification and adherence counselling essential

Referral and Red Flags

- **Urgent ophthalmology referral** if:
    - IOP ≥ 35 mmHg or uncontrolled on ≥ 2 agents
    - Rapid field loss, optic nerve haemorrhage, or disc pallor
    - Signs of lens instability or phacodonesis pre-surgery
- **Surgical referral:** trabeculectomy ± phaco; anticipate capsule rupture or IOL subluxation risk
- 💭 Reasoning: PXG eyes behave unpredictably—better to escalate early

Review of Systems / Key Associated Conditions

- Systemic associations:
    - Cardiovascular disease (hypertension, vascular fragility)
    - Alzheimer's disease (possible protein overlap)
    - Hearing loss and skin elasticity changes (fibrillin mutation pattern)
- **Genetics:** LOXL1 gene variant linked to PXG and PXF
- 💭 Reasoning: pseudoexfoliation is a systemic extracellular matrix disorder, not just ocular

Key Study – EMGT (Early Manifest Glaucoma Trial)

- Conducted in Sweden; included PXG and POAG patients
- Found that **each 1 mmHg reduction in IOP decreased progression risk by ~10%**
- **PXG** subgroup: higher baseline IOP, faster progression, more advanced visual field loss at diagnosis
- 💭 Reasoning: demonstrates importance of early detection and aggressive IOP lowering in PXG

Comparison Table – PDS/PG vs PXG

Feature	Pigment Dispersion / Pigmentary Glaucoma	Pseudoexfoliation Glaucoma
Typical age	Young (20–40 yrs)	Older (60–80 yrs)
Sex	Male > Female	Female ≥ Male
Refractive error	Myopic	Emmetropic or hyperopic
Pathophysiology	Posterior iris bowing → iris–zonule friction → pigment release	Abnormal fibrillar material from basement membrane deposits on anterior structures
Gonioscopy	Dense, uniform TM pigmentation (360°)	Patchy TM pigmentation, often inferiorly

Iris features	Mid-peripheral TIDs	Irregular, moth-eaten pupillary ruff
Lens findings	Pigment on posterior capsule	White flaky material on anterior capsule (bull's-eye pattern)
IOP behaviour	Fluctuating; spikes with exercise/dilation	High and erratic; greater diurnal variation
Risk of cataract surgery issues	Minimal	High (zonular weakness, capsular rupture)
Common demographics	Myopic, younger, male, often Asian	Older, Scandinavian/Caucasian
Disease progression	May stabilise with age	Rapid and aggressive; poor control
SLT response	Good initially, but monitor for IOP spikes	Often reduced efficacy; needs staged treatment
Systemic link	None strong	Extracellular matrix disorder (LOXL1 gene)

MICROBIAL KERATITIS (Contact Lens–related)

Case Overview

- **Patient:** 24-year-old, left eye sore (pain 5/10), watery and red for 2 days.
- **Vision:** RE 6/12, LE 6/7.5 — mild reduction.
- **Symptoms:** Watery discharge, mild photophobia, no itching, gradual onset, minimal change over 2 days.
- **History:**
    - Daily disposable contact lens wearer with reportedly good hygiene, no trauma or systemic illness.
    - Still wearing lenses during onset ➔ likely precipitating factor.
    - No prior ocular infections, no medications.
    - No swimming, no organic exposure.
    - 💭 *Reasoning:* Contact lens wear + mild AC reaction = strong suspicion for **bacterial keratitis (MK)** over sterile infiltrate.

Entrance and Anterior Findings

- **IOP:** 16 mmHg OD, 14 mmHg OS.
- **Pupils / Motilities:** Normal, no RAPD.
- **Slit Lamp:**
    - Conjunctival injection.
    - Small limbal neovascularisation.
    - Two discrete lesions consistent with **infiltrates with overlying epithelial defect (ulcers)** — likely infectious.
    - Mild anterior chamber reaction (cells and flare).

- Smooth lids.
- 💭 *Reasoning:* AC reaction → hallmark of *infectious* keratitis.
- 💭 *Tip:* If unsure between *sterile* vs *infectious*, cells/flare strongly favour **microbial keratitis**.

Diagnosis and Differentials

- **Diagnosis: Microbial keratitis (contact lens–related bacterial keratitis).**
- **Differentials:**
  - Sterile infiltrate — no AC reaction, smaller, less pain.
  - Fungal keratitis — feathery borders, satellite lesions, history of vegetative trauma.
  - Acanthamoeba keratitis — ring infiltrate, severe pain disproportionate to findings, poor lens hygiene/water exposure.
  - Herpetic keratitis — dendritic pattern, reduced corneal sensation.
- 💭 ***Reasoning:* Presence of pain, AC reaction, and overlying defect = MK.**

Management (includes Review, Prognosis, Referral, Red Flags)

- **Immediate actions:**
  - Cease **contact lens wear** until full resolution and re-epithelialisation.
  - Collect corneal scraping **prior to antibiotics** if lesion large (>2mm), deep, central, unresponsive, or atypical.
- **Empirical antibiotic therapy:**
  - **Ciprofloxacin 0.3%** — **every 15 min for 6 hrs**, then **every 30 min for remainder of day 1**.
  - **Day 2:** hourly; **then** 2-hourly until epithelial defect closes (minimum 14 days)
  - 💭 *Reasoning:* Empirical fluoroquinolone monotherapy is effective for small-to-moderate ulcers; provides broad Gram ± coverage.
- **Alternative:**
  - **Fortified antibiotic therapy** for large/central ulcers →
    - **Cefazolin 5% (Gram+) + Gentamicin 1.5% (Gram–)**, administered **alternately every 30 min–hourly**, not mixed (β-lactam inactivates aminoglycoside).
    - 💭 *Reasoning:* Fortified = hospital-based; indicated for central, deep, or sight-threatening ulcers.
- **Pain and inflammation management:**
  - **Cycloplegic:** Atropine 1% BD until AC reaction resolves (relieves ciliary spasm).
  - **Oral analgesia:** Paracetamol 500–1000 mg QID PRN.
  - 💭 *Reasoning:* Pain in MK arises from inflammation and ciliary spasm — cycloplegia + systemic analgesia addresses both.
- **Steroid consideration (SCUT trial):**
  - *Steroids for Corneal Ulcers Trial (SCUT)* — 500 participants with culture-proven ulcers.
    - All treated with **Moxifloxacin** ± **Prednisolone Sodium Phosphate 1%** after 48 hrs.
    - **Findings:**
      - Slight improvement in vision for **Pseudomonas ulcers**, possible harm for **Nocardia**.
      - Early steroids (<3 days) better than delayed.
      - No overall significant benefit; slower epithelial healing possible
      .
  - **Clinical use:** Add **Pred Forte 1% QID** *only after full epithelial closure*, if persistent stromal inflammation/scarring risk.
  - 💭 *Reasoning:* Steroids suppress inflammation but can delay healing; timing is crucial.
- **Review schedule:**
  - Review **daily** until improvement noted (pain ↓, defect smaller, staining reduced).

    - Continue frequent dosing until re-epithelialisation → then taper.
    - Once epithelialised → **every 2–3 days**, then **weekly**.
- **When to refer urgently:**
    - Lesion **>2mm**, **central**, or **with hypopyon**.
    - **Non-healing after 48 hrs** on intensive therapy.
    - **Suspected fungal/Acanthamoeba keratitis**.
    - **Vision-threatening location.**
    - **Requires fortified antibiotics / hospitalisation.**

Drug Notes and Considerations

- **Ciprofloxacin 0.3%:**
    - Forms **white crystalline deposits** on cornea — benign, indicate high local concentration.
    - Slows epithelial healing but less toxic than ofloxacin.
    - Contraindicated in **pregnancy (Category C)** and patients with **tendon damage risk** (fluoroquinolone class).
- **Ofloxacin 0.3%:**
    - Slightly faster healing but higher rate of **corneal perforation** in some studies.
    - 💭 *Reasoning:* Cipro preferred for safety; both broad-spectrum but cipro better Gram– coverage.

- **Fortified antibiotic pairs:**

EXTRA

- **Pathophysiology Overview**
    - Contact lens wear → microtrauma, epithelial hypoxia, reduced tear exchange.
    - Bacteria (e.g. **Pseudomonas aeruginosa, Staphylococcus aureus, Streptococcus pneumonia**) penetrate via epithelial defect → stromal necrosis, leukocyte infiltration, inflammatory cytokines.
    - AC reaction (cells + flare) occurs secondary to infection and inflammation.
    - 💭 *Reasoning:* Infectious keratitis destroys corneal tissue via bacterial toxins and immune-mediated inflammation; treating infection = treating inflammation.
- **Systemic and Prognostic Considerations**
    - **Expected recovery:** epithelial closure within 1–2 weeks for small ulcers.
    - **Prognosis:** good if treated early; guarded if central or large.
    - **Complications:** corneal thinning, scarring, secondary glaucoma, perforation.
    - **Systemic conditions:** diabetes, immunosuppression → poorer outcomes.
- **Review of Systems / Key Associations**
    - **Contact lens overwear** = #1 cause in developed countries.
    - **Fungal / Acanthamoeba keratitis** risk if organic trauma, poor lens hygiene, or water exposure.
    - **Post-surgery or immunocompromised** → culture early.
    - 💭 *Reasoning:* Early cultures guide targeted therapy and improve prognosis.

Table 1. Infectious Keratitis Comparison

Type	Common Pathogens	Typical Appearance	Key Risk Factors	First-line Management	Notes / Hallmarks

Bacterial Keratitis (MK)	*Staphylococcus aureus* (**Gram-positive**) *Streptococcus pneumoniae* (**Gram-positive**) *Pseudomonas aeruginosa* (**Gram-negative**) *Neisseria gonorrhoeae* (**Gram-negative**)	Focal white/grey infiltrate with overlying epithelial defect, stromal oedema, AC cells + flare	Contact lens wear, trauma, prior ocular surface disease	Topical **Ciprofloxacin 0.3%** every 15 min × 6 h → hourly; or fortified antibiotics if large/central	Rapid onset, painful, purulent, hallmark = **AC reaction**
Fungal Keratitis	*Fusarium*, *Aspergillus*, *Candida*	Feathery edges, satellite lesions, raised dry infiltrate	Vegetative trauma, topical steroids, tropical climate	Topical **Natamycin 5%** or **Amphotericin B 0.15%**, hourly initially	Slow onset, pain less severe, poor response to antibiotics
Acanthamoeba Keratitis	*Acanthamoeba castellanii*	Ring-shaped stromal infiltrate, perineural infiltrates	Water exposure, contact lens hygiene	**PHMB 0.02%** or **Chlorhexidine 0.02%** hourly	Pain **disproportionate** to clinical signs; often misdiagnosed
Viral (Herpetic) Keratitis	*HSV-1*	Dendritic ulcer with terminal bulbs	History of cold sores, stress, immunosuppression	**Acyclovir 3% ointment** 5×/day	Decreased corneal sensation, contraindication for steroids in active epithelial stage

Table 2. Fortified Antibiotic Pairing and Mechanisms

Fortified Drug	Coverage	Mechanism of Action	Notes / Key Points
Cefazolin 5%	Gram-positive	β-lactam antibiotic; inhibits bacterial cell wall synthesis	Effective for *Staph* and *Strep* species; part of standard fortified pair

Gentamicin 1.5%	Gram-negative	Aminoglycoside; inhibits 30S ribosomal protein synthesis	Excellent *Pseudomonas* coverage; potential epithelial toxicity
Tobramycin 1.3%	Gram-negative (alt to gentamicin)	Similar to gentamicin	Often used interchangeably if gentamicin unavailable

💭 **Clinical note:** Do **not mix** fortified antibiotics — β-lactam ring in cefazolin **inactivates** the aminoglycoside when combined in the same vial. Administer **alternately every 30–60 min** for central or severe ulcers.

Table 3. Quick Recall — Fluoroquinolone Monotherapy Options

Drug	Spectrum	Comments
Ciprofloxacin 0.3%	Broad (Gram ±)	Slower healing but less perforation risk; forms harmless white corneal precipitates; Category C (avoid in pregnancy)
Ofloxacin 0.3%	Broad (Gram ±)	Slightly faster healing but higher risk of perforation; Category C

💭 **Key Exam Reasoning Prompts**

- Why is an **AC reaction** the hallmark of microbial keratitis?
  ➔ Indicates infection has breached the corneal epithelium, triggering inflammatory cell migration.
- Why do we use **Ciprofloxacin over Ofloxacin**?
  ➔ Less risk of perforation; though slower healing, safer for extended therapy.
- When do we **add steroids**?
  ➔ After full epithelial healing if persistent stromal inflammation; refer to **SCUT trial** for evidence basis.
- Why can't we mix fortified antibiotics?
  ➔ β-lactam (cefazolin) chemically deactivates aminoglycoside (gentamicin).
- When to **refer immediately**?
  ➔ Central, >2 mm, deep ulcers, hypopyon, poor healing after 48 h, or uncertain aetiology.
- 💭 *Tip:* Remember — Gram-positive organisms have a **thick peptidoglycan wall** (stain purple), while Gram-negative organisms have a **thin wall and an outer membrane** (stain pink/red).

Acute Primary Angle-Closure Glaucoma (APACG)

- Case overview
  - 67-year-old Chinese female with 7–9 days of temporal flashes and increasing floaters; severe painful red eye after accidental dilation at hospital (IOP spiked to 60 mmHg).
  - HLA-B27 positive with past uveitis; high cholesterol; strong family history of retinal detachment (both parents).
  - Vision at presentation: R 6/18, L 6/15; pinhole worse due to dense cataracts.
  - IOP initially R 24, L 22; Van Herick unreliable due to arcus → gonioscopy indicated.
  - B-scan (clinic and hospital) showed no retinal tear/detachment.
  - Key APACG episode features: corneal oedema/haze, fixed mid-dilated pupil, severe pain with nausea/vomiting.
- History questions
  - Pain/onset: timing, severity, intermittent episodes in dim light or after dilation.
  - Vision: halos, blur, transient obscurations.
  - Systemic/medications: pregnancy; asthma/COPD; cardiac disease/bradycardia; renal/hepatic disease; MAOI use; topiramate/anticholinergics/TCAs.
  - Ocular: prior angle symptoms, uveitis/herpetic disease, prior narrow-angle advice or LPI, fellow-eye symptoms.
  - Risk profile: Asian ethnicity, female, hyperopia/short axial length, thick lens/cataract.
  - 💭 Prompt: always state “I will check contraindications before instilling any IOP-lowering drops.”
- Symptoms (subjective)
  - Severe ocular pain radiating to temple; headache; nausea/vomiting (often projectile).
  - Blurred vision with halos; photophobia less marked than in anterior uveitis.
  - Episodes precipitated in dim light or by mydriasis.

Testings

- Entrance tests (VA, IOP, motilities, pupils)
  - VA reduced; pinhole may worsen with dense cataracts.
  - IOP often >40–50 mmHg during acute attack.
  - EOMs normal.
  - Pupil mid-dilated, sluggish/fixed in affected eye; fellow eye normal.
  - Van Herick as able, but gonioscopy required when arcus limits view (do not skip).
  - 💭 Prompt: if posterior view limited, plan B-scan before dilation.
- Slit-lamp / anterior segment
  - Conjunctival injection; microcystic corneal oedema; shallow anterior chamber.
  - Iris bombe in pupillary-block cases; possible peripheral anterior synechiae if chronic.
  - Lens: cataract; look for glaukomflecken (after repeated attacks).
  - Mild AC reaction can occur (avoid confusing with primary uveitis).
  - Gonioscopy: appositional or synechial closure; confirm mechanism when cornea clears.
- Posterior segment (as relevant)
  - If view limited, perform B-scan (rule out retinal tear/detachment).
  - After IOP control and cornea clears, dilate cautiously for fundus exam.
- Diagnosis and differentials
  - Primary diagnosis: APACG precipitated by pharmacologic/mydriatic dilation in anatomically narrow angles.
  - Key differentials for a painful red eye with high IOP:

        - Acute anterior uveitis: dull ache, photophobia, miotic pupil, KPs/cells/flare; IOP normal/low or mildly raised.
        - Scleritis: deep boring pain, violaceous hue, normal/low IOP; systemic associations.
        - Giant cell arteritis: scalp/temporal tenderness, jaw claudication, sudden vision loss; IOP not characteristically high.
        - Neovascular glaucoma: rubeosis, secondary angle closure after CRVO/DR; often very high IOP.
        - Chronic angle closure: recurrent mild attacks, PAS, optic nerve change.
        - Posner–Schlossman (glaucomatocyclitic crisis): unilateral IOP spikes with minimal inflammation.
    - 💭 **Prompt: when uncertain, contrast pain quality, pupil shape, AC reaction, and IOP pattern.**

Treatment

- **Management, review/prognosis/tapering, referral/red flags (grouped with deep indents)**
    - **Immediate first-aid (before definitive treatment)**
        - Check contraindications first (pregnancy, asthma/COPD, bradycardia/heart block, renal/hepatic disease, MAOIs, sulfonamide allergy).
        - Instil one drop each, 2–5 minutes apart (avoid prostaglandin analogues acutely):
            - Timolol 0.5%: avoid asthma/COPD, bradycardia/heart block, hypotension.
            - Brimonidine 0.2%: avoid with MAOIs; caution severe depression.
            - Dorzolamide 2% or brinzolamide 1%: avoid significant renal failure/sulfa allergy.
        - Systemic: acetazolamide 500 mg PO stat (avoid sulfa allergy, renal/hepatic failure, pregnancy).
        - Pilocarpine 1–2% only after IOP <40 mmHg:
            - Rationale: at very high IOP the iris sphincter is ischaemic/paralysed → pilocarpine ineffective/worsens block.
        - If refractory or vision-threatening corneal oedema: mannitol 20% IV (with ophthalmology).
        - Analgesia/anti-emetic as needed.
        - 💭 Prompt: be ready to say exactly which drugs you would hold if the patient has asthma, CKD, pregnancy, or MAOI use.
    - **Definitive treatment (once cornea clears)**
        - Laser peripheral iridotomy (LPI) in the affected eye; prophylactic LPI in fellow eye.
        - If plateau iris suspected after LPI: consider argon laser peripheral iridoplasty.
        - Lens-driven (phacomorphic) component: cataract surgery when stable.
        - 💭 Prompt: explain why LPI addresses pupillary block but not plateau iris (ciliary body configuration).
    - **Review and prognosis**
        - Recheck IOP, corneal clarity, angle status within hours; then 1–2 days; then weekly until stable.
        - Monitor for optic nerve/RNFL change and visual field once media clear.
        - Counsel on recurrence risk and precipitating factors (dim light/mydriasis).
    - **Tapering/adjuncts**
        - After acute control, taper topical agents based on IOP and angle status post-LPI.
        - Short course topical steroid may be used for post-LPI inflammation (check IOP).
        - Avoid pilocarpine chronically unless specifically indicated; review side effects.
    - **Referral and red flags**
        - Urgent same-day ophthalmology if IOP remains elevated, cornea too hazy for LPI, or atypical/secondary mechanism suspected.

- Emergency referral if vision rapidly declining, severe corneal oedema, refractory pain/nausea.
- 💭 Prompt: state your escalation pathway clearly (who you call and why).

Mechanism of ACC

Table 1 – Pupil Appearance and Typical Glaucoma/Anterior Segment Conditions

Pupil appearance	Typical scenario	Key notes
Mid-dilated, sluggish/fixed	**Acute primary angle closure**	Ischaemic iris sphincter; poor light reaction; corneal oedema present
Miotic, photophobic	**Acute anterior uveitis**	Ciliary spasm causes pain and small pupil; KPs and cells/flare visible
Irregular, distorted	**Posterior synechiae (chronic uveitis)**	Adhesions between iris and lens; partial dilation only
Large, non-reactive	**Pharmacologic dilation / trauma**	Mydriatic drops or sphincter damage; distinguish from APACG via clear cornea and normal IOP
Rubeotic (poorly reactive, red iris vessels)	**Neovascular glaucoma**	Rubeosis from CRVO, diabetic retinopathy; secondary angle closure likely
Fixed, dilated with glaukomflecken	**After repeated APACG attacks**	Lens epithelial necrosis; chronic damage marker

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Table 2 – Angle-Closure Mechanisms and Typical Features

Mechanism	Pathophysiology / key sign	Management focus
Pupillary block	Aqueous trapped behind iris → iris bombe → angle obstruction	Laser peripheral iridotomy (LPI) to bypass block
Plateau iris	Anteriorly positioned ciliary processes hold iris forward; angle narrow despite patent PI	Gonioscopy/indentation required; argon laser peripheral iridoplasty if persistent
Phacomorphic	Thickened cataractous lens crowds anterior chamber and narrows angle	Manage IOP acutely, then cataract extraction
Lens subluxation / phacotopic	Lens displacement pushes iris root forward	Surgical reposition/removal after IOP control

Neovascular (secondary pull)	Fibrovascular membrane from CRVO/DR “pulls” angle closed	Pan-retinal photocoagulation + anti-VEGF + IOP control
Inflammatory / ciliary body effusion (secondary push)	Choroidal/ciliary swelling pushes iris-lens diaphragm forward	Treat underlying inflammation; cycloplegics; avoid miotics

- **First-aid checklist (rapid recall)**
  - Confirm no retinal detachment with B-scan if view limited.
  - Check contraindications to each medication before use.
  - Instil beta-blocker + alpha-agonist + CAI; give acetazolamide PO if safe.
  - Do not use pilocarpine until IOP <40 mmHg.
  - Arrange urgent ophthalmology for LPI; prophylactic fellow-eye LPI.
  - Document patient education (symptoms, triggers, when to present urgently).
  - 💭 Prompt: if every usual agent is contraindicated (e.g., pregnancy + asthma + CKD + MAOI), state that you will contact ophthalmology/GP immediately for alternative strategy (e.g., IV mannitol, urgent LPI).
